

Apriori: Implementation Tricks

□ How to generate candidates?

□ Step 1: self-joining F_k

□ Step 2: pruning

□ Example of candidate-generation

□ $F_3 = \{abc, abd, acd, ace, bcd\}$

□ Self-joining: $F_3 * F_3$

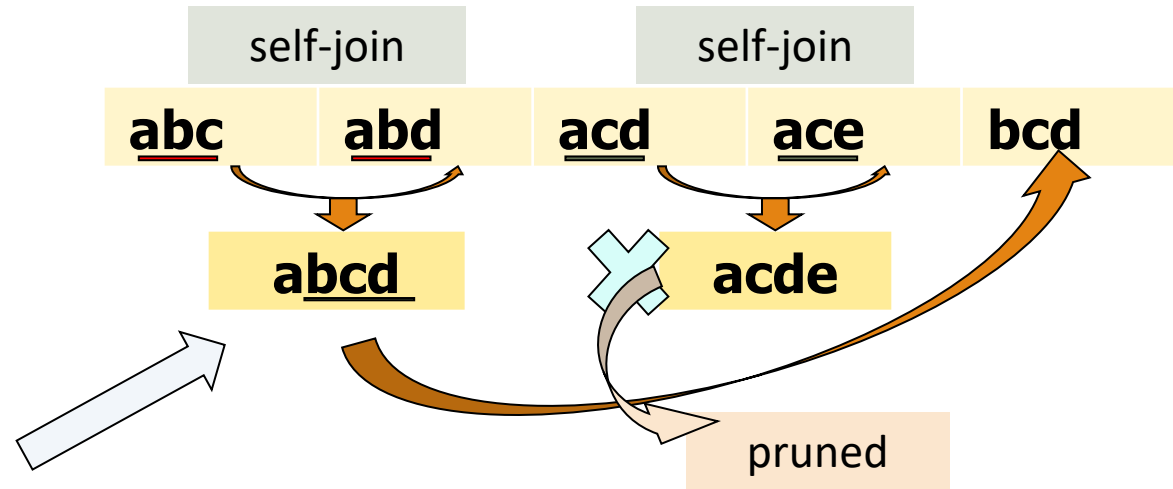
□ $abcd$ from abc and abd

□ $acde$ from acd and ace

□ Pruning:

□ $acde$ is removed because ade is not in F_3

□ $C_4 = \{abcd\}$



Candidate Generation (Pseudo-Code)

□ Suppose the items in F_{k-1} are listed in an order

□ // Step 1: Joining

for each p in F_{k-1}

for each q in F_{k-1}

if $p.item_1 = q.item_1, \dots, p.item_{k-2} = q.item_{k-2}, p.item_{k-1} < q.item_{k-1}$ {

$c = \text{join}(p, q)$

□ // Step 2: pruning

if $\text{has_infrequent_subset}(c, F_{k-1})$

continue // prune

else add c to C_k

}

